Kerui Wu

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Education

University of Massachusetts Amherst (UMass Amherst) Sept. 2025 – Present
Ph.D. in Computer Science, GPA: 3.90/4.0
Advisor: Prof. Mingda Qiao

Rensselaer Polytechnic Institute (RPI) Sept. 2021 – May 2025
B.S. in Computer Science, GPA: 3.76/4.0 *Magna Cum Laude*

Research Focus

- **Learning Systems** (Reinforcement Learning, Online Learning)
- **AI Agents** (Sequential Decision-Making, Multi-Agent Coordination, Resource Allocation)

Publications

(* Co-first authors; $(\alpha - \beta)$ alphabetic order)

- **Adversarially Batched Online Learning**
  - $(\alpha - \beta)$ Mingda Qiao, Kerui Wu
  - Under Submission
- **Compute Allocation for LLM-based Evolutionary Search: From Depth-Breadth to Bandits**
  - Kerui Wu*, Sixue Xing*, Haoyu He*, Zhuo Yang, Haozheng Luo, Tianfan Fu, Aarthy Nagarajan
  - **EMNLP 2026** (Acceptance Rate: 15%) — **Oral**
- **ClinicalReTrial: Clinical Trial Redesign with Self-Evolving Agents**
  - Sixue Xing, Kerui Wu, Xuanye Xia, Meng Jiang, Jintai Chen, Tianfan Fu
  - **EMNLP 2026** , Findings (Acceptance Rate: 29%)
- **On the Adversarial Robustness of Graph Neural Networks with Graph Reduction**
  - Kerui Wu, Ka-Ho Chow, Wenqi Wei, Lei Yu
  - **ESORICS 2025** (Acceptance Rate: 16%) — **Oral**
- **Large EEG-U-Transformer for Time-Step level Detection**
  - Kerui Wu, Ziyue Zhao, Bülent Yener
  - Technical Report, **arXiv** — **Winning Model of 2025 Seizure Detection Challenge**
- **On the Robustness of Graph Reduction Against GNN Backdoor Attacks**
  - Yuxuan Zhu, Michael Mandulak, Kerui Wu, George Slota, Yuseok Jeon, Ka-Ho Chow, Lei Yu
  - **AISec@CCS 2024** (Acceptance Rate: 25%)

Awards

NSF NAIRR Pilot Start-Up Award, PI 2026
National Artificial Intelligence Research Resource (NAIRR)
CICS Scholarship 2025
Manning College of Information & Computer Sciences, UMass Amherst
Best Algorithm Award 2025
2025 Seizure Detection Challenge, International Conference on AI in Epilepsy
Certified in CyberSecurity (CC) 2023
ISC² (International Information System Security Certification Consortium)

Research Experience

University of Massachusetts Amherst (UMass Amherst)
Sept. 2025 – Present
Advisor: Mingda Qiao *Ph.D. Research Assistant*

Robust Reinforcement Learning & Online Learning

- Characterize **RLVR training dynamics** under **LLM-as-a-judge** and reward models with systematic verification errors, modeling noise effects on the training trajectory.
- Benchmark **noise-corrected reinforcement learning** across reward-noise regimes and develop `ROM`, a robust RL algorithm that recovers **clean-reward GRPO performance** under all evaluated noisy settings.

- Establish theoretical foundations for **adversarially batched online learning**, where policies can be updated only under restricted, adversarial update schedules.
- Design **provably optimal learning algorithms**, including a generic **adaptive-doubling scheme** that requires no prior knowledge of update times while preserving optimal performance guarantees.

Multi-Agents with Sequential Decision-Making

- Systematically characterize compute allocation in **LLM-guided self-evolution**, uncovering a fitness–compute envelope and **capability-dependent depth–breadth trade-offs** across models and tasks.
- Develop `BaSE`, an adaptive **Multi-Armed Bandits**-based framework that dynamically allocates LLM calls from stagnating runs to promising trajectories, achieving *12.3%* fitness improvement across five LLMs (e.g., **Qwen3** and **Llama**) over three tasks.
- Build `ClinicalReTrial`, a reasoning-driven **LLM multi-agent system** that casts clinical trial redesign as an iterative protocol optimization problem.
- Combine **retrieval-augmented generation (RAG)**, **hierarchical memory**, and **in-context learning** for continual agent self-refinement, achieving an *83.3%* protocol-improvement rate and *5.7%* success probability improvement.

Rensselaer Polytechnic Institute (RPI)
Jan. 2024 – May 2025

Advisor: Lei Yu, Bülent Yener — *Undergraduate Researcher*

Efficient Representation Learning for Long-Sequence EEG

- Introduce a time-step-level **sequence modeling formulation** for long-sequence EEG, eliminating redundant sliding-window inference and enabling **event-centric prediction**.
- Design a scalable **Conv-Transformer architecture** with multi-scale CNN encoders, residual convolutional blocks, and **self-attention temporal embeddings**.
- Achieve a **10× runtime speedup** and a *19.4%* F1-score improvement, winning the **1st Place** in the 2025 Seizure Detection Challenge at the International Conference on AI in Epilepsy.

Adversarial Robustness of Graph Neural Networks

- Empirically study the **robustness of Graph Neural Networks** against poisoning, evasion, and backdoor attacks.
- Analyze the security implications of graph reduction for **scalable GNNs**, characterizing how structural compression changes the impact of adversarial perturbations.

Selected Projects

China Construction Bank Achievement System — React, Go, MySQL, Azure

- Developed a production web platform for tracking employee performance, **deployed across 11 branches of China Construction Bank.**
- Implemented full-stack functionality, including data dashboards, analytics queries, and role-based management tools.
- Designed secure authentication and API infrastructure using **SHA256 hashing and JWT**, and deployed the system on **Microsoft Azure cloud**.

Submitty Open-Source Contribution — PHP, JavaScript, SQL, Python

- Contributed to Submitty, an open-source **course management** and **automated grading** system adopted by multiple universities.
- Implemented **full-stack features** including customizable pronoun settings across frontend UI, backend APIs (PHP), and relational database schemas (SQL). **5 pull requests were merged into the main branch of the repository.**

Technical Skills

Machine Learning: PyTorch, Torch-Geometric, Scikit-Learn, vLLM, LangChain, MCP
Programming: Python, C/C++, JavaScript, Go, PHP, SQL

Services

Program Committee Member: ICLR 2027